

A rational completely positive matrix with no rational nonnegative factor

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Abstract

An explicit integer matrix of order 444 and rank seven is constructed that is completely positive over the reals but has no nonnegative rational Gram factor of any finite width. All its entries are strictly positive, and its real cp-rank is seven. The construction gives a negative answer to the universal boundary-factorization question PF-03. The obstruction is a rational polyhedral cone in $\mathbb{R}^7$ on which a trace-zero quadratic form is nonnegative and has exactly seven irrational zero rays forming an orthonormal basis. The arithmetic seed lies in $\mathbb{Q}(\sqrt[3]{2})$. Rational interval calculations certify its signs, and exhaustive integer computation certifies all 444 facets of the cone. The complete matrix, a compact integer representation, and a standard-library Python verifier accompany the proof. This is a computer-assisted proof that passed an independent informal Codex AI-agent audit on 13 September 2026; the accompanying independent-review report records the checks. It has not received external human peer review or formal proof-assistant verification.

1 The statement being refuted

A real symmetric matrix A is completely positive if $A = BB^\top$ for some entrywise nonnegative real matrix B with finitely many columns. Write $\mathcal{CP}_n$ for the cone of such matrices of order n . The question in PF-03 is whether every

$$A \in \mathbb{Q}^{n \times n} \cap \partial \mathcal{CP}_n, \quad n \geq 5, \quad (1.1)$$

admits $A = CC^\top$ with $C \in \mathbb{Q}_{\geq 0}^{n \times m}$ for some finite m . The width is explicitly unrestricted. Boundaries are in the full space of real symmetric matrices.¹

The distinction between a rational Gram factor and a *nonnegative* rational Gram factor is essential. Likewise, failure of a square rational factor does not suffice: a wider rational factor could still exist. The argument below excludes every finite width. The interior theorem and the boundary question are stated separately in the 2025 survey of Berman and Shaked-Monderer.²

Theorem 1.1 (Explicit counterexample). *Let $R \in \mathbb{Z}^{444 \times 7}$ be the primitive integer facet matrix defined in Sections 4–5, in the row order specified there, and set*

$$A = RR^\top. \quad (1.2)$$

Then all entries of A are strictly positive integers,

$$\text{rank}(A) = \text{cpr}(A) = 7, \quad A \in \partial \mathcal{CP}_{444},$$

and there is no $C \in \mathbb{Q}_{\geq 0}^{444 \times m}$ satisfying $CC^\top = A$, for any finite integer $m \geq 1$.

¹See the exact PF-03 statement in [1], especially its unrestricted-width qualification.

²[2], Section 2, Question 2.1, Theorem 2.2, and the boundary Open Problem on page 2. The interior theorem is also [3], Theorem 1.1.

The exact 444×7 array is supplied as `data/R_integer.csv`. The expanded lower triangle of A is supplied as `data/A_integer.mtx.gz`. Thus the matrix is available both explicitly and in a compact exact representation. Equation (1.2) is *not* its nonnegative factorization: R is a signed integer matrix. Its nonnegative real factor is RO , where O is constructed below.

2 A general unrestricted-width obstruction

For a cone $K \subset \mathbb{R}^r$, write

$$\mathcal{CP}(K) = \text{cone}\{xx^\top : x \in K\}.$$

All conic combinations here are finite. The following lemma isolates the part of the argument that excludes arbitrary factor widths.

Lemma 2.1 (Cone obstruction). *Suppose that $K \subset \mathbb{R}^r$ is a full-dimensional pointed rational polyhedral cone and*

$$K = \{x \in \mathbb{R}^r : Rx \geq 0\}, \quad R \in \mathbb{Q}^{N \times r}, \quad \text{rank } R = r.$$

Suppose $O = [u_1 \ \cdots \ u_r] \in \mathbb{R}^{r \times r}$ is orthogonal and every $u_i \in K$. Suppose also that a real symmetric matrix Q satisfies

$$\text{tr } Q = 0, \quad x^\top Qx \geq 0 \ (x \in K), \quad \{x \in K \cap \mathbb{Q}^r : x^\top Qx = 0\} = \{0\}. \quad (2.1)$$

Then $A = RR^\top$ is completely positive and admits no nonnegative rational Gram factor of any finite width. If $N > r$, it is a boundary matrix of $\mathcal{CP}_N$.

Proof. Since $RO \geq 0$ and $OO^\top = I_r$,

$$A = (RO)(RO)^\top \in \mathcal{CP}_N. \quad (2.2)$$

Assume, for a contradiction, that $A = CC^\top$ with a finite nonnegative rational matrix $C = [c_1 \ \cdots \ c_m]$.

Every column of C lies in range R . Indeed, if $y \in \ker R^\top$, then

$$0 = y^\top Ay = \sum_{j=1}^m (y^\top c_j)^2.$$

All summands are nonnegative real numbers, so $y^\top c_j = 0$ for every j . Taking the orthogonal complement of $\ker R^\top$ gives the assertion.

The matrix

$$L = (R^\top R)^{-1} R^\top$$

is rational and satisfies $LR = I_r$. Put $X = LC = [x_1 \ \cdots \ x_m]$. Then X is rational, $RX = C \geq 0$, and

$$XX^\top = LAL^\top = I_r. \quad (2.3)$$

In particular every $x_j \in K$. Now

$$0 = \text{tr } Q = \text{tr}(QXX^\top) = \sum_{j=1}^m x_j^\top Qx_j. \quad (2.4)$$

Every summand is nonnegative. Each must vanish, and (2.1) forces every $x_j = 0$, contradicting (2.3). No upper bound or other restriction on m has been used.

If $N > r$, then A is singular. For a nonzero $z \in \ker A$, the matrix $A - \varepsilon z z^\top$ fails to be positive semidefinite for every $\varepsilon > 0$. Since every completely positive matrix is positive semidefinite, no neighborhood of A is contained in $\mathcal{CP}_N$. Together with $A \in \mathcal{CP}_N$, this places A on its boundary. $\square$

We will certify a stronger zero-set property than the one in (2.1): the nonzero zeros in K are exactly the positive rays through the orthonormal vectors $u_1, \dots, u_7$, and none of these rays contains a nonzero rational vector.

Lemma 2.2 (Irrational ray test). *Let $\alpha = \sqrt[3]{2}$ and*

$$u = a + \alpha b + \alpha^2 c, \quad a, b, c \in \mathbb{Q}^r.$$

If $\text{rank}[a \ b \ c] > 1$, then $\mathbb{R}u$ contains no nonzero rational vector.

Proof. The polynomial $t^3 - 2$ is irreducible over $\mathbb{Q}$ by Eisenstein's criterion, so $1, \alpha, \alpha^2$ are linearly independent over $\mathbb{Q}$. Suppose $u = sw$ with $0 \neq w \in \mathbb{Q}^r$. Choose a nonzero coordinate w_i . Then $s = u_i/w_i \in \mathbb{Q}(\alpha)$, so write $s = s_0 + s_1\alpha + s_2\alpha^2$ with $s_k \in \mathbb{Q}$. Comparing coefficients gives $a = s_0w$, $b = s_1w$, and $c = s_2w$. The coefficient matrix has rank at most one, a contradiction. $\square$

3 An exact seven-dimensional algebraic seed

Set $\alpha = \sqrt[3]{2} > 0$ and $\mathbb{F} = \mathbb{Q}(\alpha)$. The field is viewed in its unique real embedding. Define the following skew-symmetric integer matrices:

$$S = \begin{pmatrix} 0 & -1 & 0 & -6 & -2 & -6 & -4 \\ 1 & 0 & 3 & 4 & -1 & 5 & -1 \\ 0 & -3 & 0 & -4 & 1 & -2 & -5 \\ 6 & -4 & 4 & 0 & 1 & 3 & -3 \\ 2 & 1 & -1 & -1 & 0 & -1 & -1 \\ 6 & -5 & 2 & -3 & 1 & 0 & -3 \\ 4 & 1 & 5 & 3 & 1 & 3 & 0 \end{pmatrix}, \quad (3.1)$$

$$T = \begin{pmatrix} 0 & 0 & -3 & 0 & -2 & 2 & 5 \\ 0 & 0 & -4 & 2 & 2 & -2 & 2 \\ 3 & 4 & 0 & 1 & 1 & -3 & 0 \\ 0 & -2 & -1 & 0 & 4 & 3 & 0 \\ 2 & -2 & -1 & -4 & 0 & -2 & -5 \\ -2 & 2 & 3 & -3 & 2 & 0 & 4 \\ -5 & -2 & 0 & 0 & 5 & -4 & 0 \end{pmatrix}. \quad (3.2)$$

Put

$$\Omega = \alpha S + \alpha^2 T, \quad O = (I_7 + \Omega)^{-1}(I_7 - \Omega). \quad (3.3)$$

The real skew-symmetry of Ω makes $I_7 \pm \Omega$ invertible. The Cayley transform is orthogonal, and all entries of O belong to $\mathbb{F}$. These identities are also checked directly in exact field arithmetic.

Expand O uniquely as

$$O = B_0 + \alpha B_1 + \alpha^2 B_2, \quad B_k \in \mathbb{Q}^{7 \times 7}. \quad (3.4)$$

For $1 \leq i \leq 7$, define

$$C_i = [(B_0)_{:i} \ (B_1)_{:i} \ (B_2)_{:i}], \quad U_i = \text{range } C_i, \quad v = (1, \alpha, \alpha^2)^\top, \quad u_i = C_i v = O_{:i}. \quad (3.5)$$

The exact matrices C_i all have rank three. Lemma 2.2 therefore applies to each u_i .

3.1 Definition of the exposing quadratic form

Let $Q' = (q_{ij}) \in \mathbb{F}^{7 \times 7}$ be symmetric with zero diagonal. Its 21 upper-triangular entries are ordered lexicographically by (i, j) . Fix the last seven entries by

$$\begin{array}{c|ccccccc} (i, j) & (3, 7) & (4, 5) & (4, 6) & (4, 7) & (5, 6) & (5, 7) & (6, 7) \\ \hline 1000q_{ij} & 63 & 1007 & 18 & 402 & 151 & 1522 & 1640. \end{array} \quad (3.6)$$

Define the remaining 14 entries by the 14 equations

$$(Q' O^\top B_k)_{ii} = 0, \quad 1 \leq i \leq 7, \quad k = 0, 1. \quad (3.7)$$

The coefficient matrix has exact rank 14, with pivots in the first 14 lexicographic upper-triangular variables. Thus these equations and (3.6) uniquely specify Q' . Finally set

$$Q = O Q' O^\top, \quad H_i = C_i^\top Q C_i. \quad (3.8)$$

This defines the entire seed using only two small integer matrices, seven rational numbers, and arithmetic in a specified cubic field. The files `build_seed.py` and `exact_algebraic_certificate.json` provide, respectively, a deterministic reconstruction and all expanded field elements. Each field element is stored as its three rational coefficients in the basis $(1, \alpha, \alpha^2)$.

Lemma 3.1 (Certified seed properties). *For the preceding exact seed,*

$$O O^\top = O^\top O = I_7, \quad \text{tr } Q = 0, \quad q_{ij} > \frac{17391}{10^6} > 0 \quad (i \neq j). \quad (3.9)$$

Every C_i has rank three, and every H_i is positive semidefinite of rank two with kernel $\mathbb{R}v$.

Proof. Orthogonality follows from the Cayley transform; cyclic invariance of trace and the zero diagonal of Q' give $\text{tr } Q = 0$. The exact coefficient computations give $\text{rank } C_i = 3$ for all seven indices. Equations (3.7), together with

$$O^\top B_0 + \alpha O^\top B_1 + \alpha^2 O^\top B_2 = I_7,$$

also imply $(Q' O^\top B_2)_{ii} = 0$. Consequently $C_i^\top Q u_i = 0$, or equivalently $H_i v = 0$.

The remaining assertions are finite sign checks in $\mathbb{F}$. Table 1 records strict rational lower bounds for the two leading principal minors needed for each H_i . The bound in (3.9) is also obtained by exact rational interval evaluation. The sign algorithm and isolating interval are specified in Section 7; they use no floating-point tolerance.

For completeness, positivity of those two minors and $H_i v = 0$ really do prove the asserted semidefiniteness. Write

$$H_i = \begin{pmatrix} D & b \\ b^\top & c \end{pmatrix}, \quad v = \begin{pmatrix} w \\ \alpha^2 \end{pmatrix}, \quad w = (1, \alpha)^\top.$$

The leading 2×2 matrix D is positive definite. The kernel equation forces $b = -Dw/\alpha^2$ and $c = w^\top Dw/\alpha^4$. Thus

$$H_i = \begin{pmatrix} I_2 \\ -w^\top/\alpha^2 \end{pmatrix} D \begin{pmatrix} I_2 & -w/\alpha^2 \end{pmatrix}. \quad (3.10)$$

This is positive semidefinite of rank two, with exactly the stated kernel. The exact certificates verify $H_i v = 0$ separately as well. $\square$

Table 1: Certified strict lower bounds; every displayed decimal is an exact rational number, not a floating-point estimate. Indices inside H_i are one-based.

i	Lower bound for $(H_i)_{11}$	Lower bound for $\det(H_i)_{\{1,2\},\{1,2\}}$
1	1.016353	0.444291
2	1.150668	0.687760
3	0.380276	0.104352
4	0.242452	0.030000
5	3.486402	4.967538
6	0.424643	0.091575
7	1.071944	0.579814

4 A rational cone with seven irrational zero rays

Define the exact rationals

$$a_0 = \frac{125992104989487316477}{10^{20}}, \quad b_0 = \frac{158740105196819947475}{10^{20}}, \quad \delta = \frac{1}{10000}, \quad (4.1)$$

and the rational matrix

$$W = \begin{pmatrix} 1 & 1 & 1 \\ a_0 + \delta & a_0 & a_0 - \delta \\ b_0 & b_0 + \delta & b_0 - \delta \end{pmatrix}. \quad (4.2)$$

Its determinant is $3\delta^2 > 0$. Let w_1, w_2, w_3 be its columns. For each i , put

$$g_{ik} = C_i w_k \in \mathbb{Q}^7, \quad G_i = \text{cone}\{g_{i1}, g_{i2}, g_{i3}\}, \quad K = G_1 + \cdots + G_7. \quad (4.3)$$

Equivalently, K is the cone generated by the 21 columns of

$$V = [C_1 W \mid C_2 W \mid \cdots \mid C_7 W] \in \mathbb{Q}^{7 \times 21}. \quad (4.4)$$

These generators need not be nonnegative in their seven-dimensional coordinates. Their nonnegative embedding is obtained from the facet matrix R in the next section.

4.1 Exact inclusion and cross-pairing certificates

Put

$$d_x = \frac{\alpha - a_0}{\delta}, \quad d_y = \frac{\alpha^2 - b_0}{\delta},$$

and define

$$\lambda_1 = \frac{1 + 2d_x - d_y}{3}, \quad \lambda_2 = \frac{1 - d_x + 2d_y}{3}, \quad \lambda_3 = \frac{1 - d_x - d_y}{3}. \quad (4.5)$$

Direct calculation gives

$$\lambda_1 + \lambda_2 + \lambda_3 = 1, \quad W\lambda = v, \quad u_i = \sum_{k=1}^3 \lambda_k g_{ik}. \quad (4.6)$$

The rational interval certificate gives $\lambda_k > 33/100$ for all three indices. Hence $u_i \in \text{ri} G_i$.

All cross-block generator pairings satisfy the following stronger-than-needed strict inequality:

$$g_{ik}^\top Q g_{j\ell} > \frac{17}{1000} \quad (1 \leq i < j \leq 7, \ 1 \leq k, \ell \leq 3). \quad (4.7)$$

There are exactly $\binom{7}{2} 3^2 = 189$ such checks. They are evaluated from $V^\top Q V$ in the same exact field arithmetic, using rational lower endpoints; the smallest certified lower endpoint exceeds 0.017344.

Finally, the rational vector

$$h = \frac{1}{10^8} \begin{pmatrix} -71246841 \\ -101280383 \\ -150296097 \\ -110993458 \\ -39802169 \\ -119030527 \\ -63286873 \end{pmatrix} \quad (4.8)$$

satisfies

$$h^\top g_{ik} > \frac{999}{1000} > 0 \quad (1 \leq i \leq 7, \ 1 \leq k \leq 3). \quad (4.9)$$

This last check is purely rational. These exact values, the barycentric coefficients, and the generator matrix are supplied in `data/rational_cone_certificate.json`.

Proposition 4.1 (The zero set on the rational cone). *The cone K is pointed, rational polyhedral, and full-dimensional in $\mathbb{R}^7$. The quadratic form $q(x) = x^\top Q x$ is nonnegative on K , and*

$$\{x \in K : q(x) = 0\} = \{0\} \cup \bigcup_{i=1}^7 \{t u_i : t > 0\}. \quad (4.10)$$

In particular, its only rational zero in K is the origin.

Proof. Finite rational generation makes K rational polyhedral. The common strictly positive functional (4.9) makes it pointed: any nonzero conic combination of generators has positive h -value, so K cannot contain both a nonzero vector and its negative. By (4.6), K contains all columns of the orthogonal matrix O . They span $\mathbb{R}^7$, proving full dimension.

For $x_i \in G_i \subset U_i$, write $x_i = C_i z_i$. Lemma 3.1 gives

$$q(x_i) = z_i^\top H_i z_i \geq 0,$$

with equality precisely when $x_i \in \mathbb{R} u_i$. Because C_i is injective and every generator w_k has first coordinate one, the first coordinate of any nonzero vector in $\text{cone}\{w_1, w_2, w_3\}$ is strictly positive. Thus the only nonzero part of $\mathbb{R} u_i$ inside G_i is its positive ray.

For nonzero $x_i \in G_i$ and $x_j \in G_j$ with $i \neq j$, write each as a nonnegative combination of its three generators. Each combination has at least one positive coefficient. Expanding the bilinear pairing and using (4.7) gives

$$x_i^\top Q x_j > 0. \quad (4.11)$$

Now any $x \in K$ has some representation $x = x_1 + \cdots + x_7$ with $x_i \in G_i$, and

$$q(x) = \sum_i q(x_i) + 2 \sum_{i < j} x_i^\top Q x_j \geq 0. \quad (4.12)$$

If at least two x_i are nonzero, the right-hand side is strictly positive. If exactly one is nonzero, it vanishes precisely on that group's positive u_i -ray. The zero vector causes no exception because the common positive functional excludes a nontrivial conic sum equal to zero. This proves (4.10).

Each C_i has rational column rank three. Lemma 2.2 shows that none of these seven lines contains a nonzero rational vector. Consequently K has no nonzero rational zero of q . $\square$

5 The exact facet matrix

The passage from a rational cone in $\mathbb{R}^7$ to an ordinary nonnegative Gram problem must use a complete inequality description of the cone. Merely finding some supporting inequalities would not suffice. The following construction and exhaustive check supply that completeness.

Multiply each of the 21 columns of V by a positive rational number to obtain a primitive integer vector. Denote the resulting vectors, in the same block order, by $p_1, \dots, p_{21} \in \mathbb{Z}^7$. No column changes its ray. For every six-element subset $J \subset \{1, \dots, 21\}$, form the 6×7 matrix with rows $p_j^\top$, $j \in J$.

If the row rank is six, compute a primitive integer generator r_J of its one-dimensional nullspace. Retain it precisely when one orientation has

$$r_J^\top p_k \geq 0 \quad (1 \leq k \leq 21). \quad (5.1)$$

Use that orientation and discard duplicate normals. Sort the retained normals first by their increasing zero-index lists $\{k : r_J^\top p_k = 0\}$ and then lexicographically by the normal entries. The transposes of these normals are the rows of R .

Lemma 5.1 (Complete facet enumeration). *This exact enumeration produces 444 distinct primitive integer normals. Every corresponding zero-index list has size six, and*

$$K = \{x \in \mathbb{R}^7 : Rx \geq 0\}, \quad R \in \mathbb{Z}^{444 \times 7}, \quad \text{rank } R = 7. \quad (5.2)$$

Proof. The finite enumeration gives the counts in Table 2. Nullspaces, ranks, orientations, and all supporting signs are calculated using exact integers. In particular, the 444 retained normals vanish on six independent generators, so each defines a six-dimensional face of K : a facet.

Conversely, every facet of a full-dimensional finitely generated cone in $\mathbb{R}^7$ is generated by the input generators lying on that facet, and has linear span of dimension six. It therefore contains six independent input generators. Their index set occurs in the exhaustive enumeration, and their oriented primitive null normal is the facet normal. No facet can be missing.

A full-dimensional pointed polyhedral cone is the intersection of its facet halfspaces. One can also see this here by slicing with $h^\top x = 1$: (4.9) gives a compact six-dimensional polytope whose vertices

are among $p_k/(h^\top p_k)$. Its facet inequalities completely determine the slice, and homogenizing gives the cone description. Thus (5.2) follows. The exact rational row-rank check gives $\text{rank } R = 7$; it is also forced by pointedness of the intersection. $\square$

Table 2: Exhaustive exact facet certificate.

Quantity	Verified value
Six-generator subsets examined, $\binom{21}{6}$	54,264
Rank-deficient six-generator subsets	0
Non-supporting six-generator subsets	53,820
Distinct supporting primitive facet normals	444
Generators on each facet	6
Rank of the complete facet matrix	7

The file `data/facets.json` records both the oriented primitive normals and their exact zero-index lists. The verifier does not trust a stored “complete” flag: it independently recomputes all 54,264 nullspaces and compares the resulting complete set with the stored one. The construction of R involves no numerical convex-hull oracle.

6 Proof of the counterexample and further consequences

Proof of Theorem 1.1. Apply Lemma 2.1 with $r = 7$, $N = 444$, the cone from (4.3), the orthogonal matrix (3.3), and the form (3.8). Lemma 3.1 gives $\text{tr } Q = 0$, Proposition 4.1 gives the required absence of nonzero rational zeros, and Lemma 5.1 gives the exact rational cone embedding. Therefore $A = RR^\top$ is completely positive, has no rational nonnegative factor of any finite width, and lies on the boundary because $444 > 7$.

It remains to verify the claimed strict positivity and cp-rank. Let $B = RO$. Every row of R is nonnegative on every generator g_{ik} . Since all three coefficients in (4.6) are positive,

$$(Ru_i)_a = 0 \iff \text{row } a \text{ of } R \text{ vanishes on all three } g_{i1}, g_{i2}, g_{i3}. \quad (6.1)$$

A facet contains exactly six of the 21 generator rays. Hence a row of B can vanish in at most two of its seven coordinates. Every row of B therefore has at least five strictly positive coordinates. Any two rows have a common strictly positive coordinate, since two subsets of size at least five in a seven-element set intersect. Their inner product is strictly positive. Thus every entry of $A = BB^\top = RR^\top$ is positive; it is an integer because R is an integer matrix.

Finally, $\text{rank } A = \text{rank } R = 7$. The displayed nonnegative factor $B = RO$ has seven columns, so $\text{cpr}(A) \leq 7$. Every Gram factor has at least $\text{rank } A$ columns, giving equality. $\square$

6.1 Why adding columns cannot repair rationality

The obstruction actually determines all possible factor directions, not just a selected factorization. If $A = CC^\top$ is any real nonnegative factorization, the same reduction $X = LC$ gives $XX^\top = I_7$ and $x_j \in K$. The trace calculation forces every nonzero x_j onto one of the seven rays in (4.10). Thus every nonzero column of C lies on one of

$$\mathbb{R}_{>0}Ru_1, \dots, \mathbb{R}_{>0}Ru_7. \quad (6.2)$$

Each ambient ray also has no nonzero rational vector: applying the rational left inverse L to such a vector would give a nonzero rational vector on $\mathbb{R}u_i$. Splitting a column into more columns along its ray cannot avoid this obstruction.

In reduced coordinates, orthogonality makes the remaining freedom explicit. Writing each nonzero $x_j = t_j u_{i(j)}$ with $t_j > 0$, the identity $XX^\top = I_7$ is equivalent to

$$\sum_{j:i(j)=i} t_j^2 = 1 \quad (1 \leq i \leq 7).$$

There is freedom to split each unit weight, but none of the permitted nonzero directions is rational.

Corollary 6.1 (Larger orders). *For every $n \geq 444$, there is an entrywise strictly positive integer matrix of order n and rank seven that is completely positive but has no rational nonnegative factor of any finite width.*

Proof. Append copies of one row of R to obtain $R_n \in \mathbb{Z}^{n \times 7}$ and set $A_n = R_n R_n^\top$. The factor $R_n O$ is nonnegative. All its row inner products are positive by the preceding support argument. A rational nonnegative factor of A_n would restrict on its first 444 rows to one of A , which is impossible. $\square$

Corollary 6.2 (Nonclosedness of rational factorability). *For order 444, the set of matrices with a finite nonnegative rational Gram factor is not closed even relative to $\mathcal{CP}_{444} \cap \mathbb{Q}^{444 \times 444}$ with its Euclidean subspace topology.*

Proof. Approximate every entry of the finite nonnegative real factor RO by nonnegative rational numbers. The Gram matrices of the resulting rational factors converge to A , but Theorem 1.1 excludes A itself. $\square$

7 Exact arithmetic, finite verification, and reproducibility

The role of computation is restricted to explicit finite equalities, ranks, signs, and an exhaustive facet list. The quantified conclusion about all rational factors is the argument in Lemma 2.1; no enumeration of candidate factor widths or rational matrices is involved.

7.1 Cubic field arithmetic and sign certificates

Represent an element of $\mathbb{F}$ by a triple of rational numbers (a, b, c) meaning $a + b\alpha + c\alpha^2$. Addition is coordinatewise, and multiplication is the exact rule

$$\begin{aligned} (a + b\alpha + c\alpha^2)(d + e\alpha + f\alpha^2) \\ = (ad + 2bf + 2ce) + (ae + bd + 2cf)\alpha + (af + be + cd)\alpha^2. \end{aligned} \quad (7.1)$$

Every rational is stored as an arbitrary-precision integer numerator and positive integer denominator. Gaussian elimination uses exact inverses in this cubic field; no algebraic-number approximations enter an equality or rank test.

For inequalities, the verifier uses

$$\begin{aligned} \ell &= \frac{12599210498948731647672106072782283505702}{10^{40}}, \\ u &= \frac{12599210498948731647672106072782283505703}{10^{40}}. \end{aligned} \quad (7.2)$$

It checks $\ell^3 < 2 < u^3$ by integer arithmetic. The real cube function is strictly increasing, so $\ell < \alpha < u$. Each quadratic expression $a + b\alpha + c\alpha^2$ is evaluated by interval Horner arithmetic on $[\ell, u]$. Products use the minimum and maximum of the four endpoint products, and all endpoints remain rational. A positivity assertion is accepted only if its exact lower endpoint is positive.

The same algorithm certifies the principal minors of H_i , all 21 off-diagonal entries of Q' , all 189 cross-generator pairings, and the barycentric coefficients. Dependency overestimation in interval arithmetic can only make the bounds wider; it cannot invalidate a positive lower endpoint.

7.2 Facet arithmetic

For each 6×7 integer matrix, the nullspace routine uses fraction-free Bareiss elimination with row and column pivoting. Every integer division is checked for zero remainder. Integer back substitution produces a null vector, which is checked against all six input rows and normalized by the positive gcd of its entries. Orienting and testing it against all 21 generators uses integer dot products only.

Both the original exact symbolic construction and the independent standard-library reconstruction produce identical algebraic certificate entries. The final distributed verifier requires no third-party packages: it rebuilds the seed from (3.1), (3.2), and (3.6), rechecks the cone, and exhaustively reconstructs the facet list. The mathematical proof does not depend on the numerical search that first suggested the seed.

7.3 Complete verification command

From the extracted package directory, run

```
python code/verify_all.py
python code/test_arithmetic.py
```

The first command verifies the seed, all cone properties, all 54,264 facet candidates, and every entry of the expanded matrix. The second runs 18 arithmetic and corruption-detection unit tests. The recorded runs passed. Python's optimization switches `-O` and `-OO` are explicitly rejected so that assertion checks cannot be disabled silently.

The verified matrix has 98,790 stored lower-triangular entries, each a strictly positive integer. The largest entry has 270 decimal digits. The factor RO has 2,966 strictly positive entries and 142 exact zeros. The Gram identity is certified by $OO^T = I_7$ and the exact definition $A = RR^T$, and the expanded matrix file is checked entry by entry against that definition.

The file `logs/verification_summary.json` records exact rational bounds and the verified counts. Unit-test results are in `logs/unit_tests.txt`. These logs are not accepted as inputs to establish the theorem: running the verifier recomputes the checks.

7.4 What is and is not being claimed

The construction refutes the universal assertion in (1.1). It does not assert that order 444 or rank seven is minimal. It does not classify every matrix of order five or the intermediate ranks. Those additional classification questions are not necessary to answer the stated universal existence question negatively.

The counterexample is singular but entrywise strictly positive. Singularity is allowed by the stated boundary problem, and the unrestricted-width condition has been addressed directly. The

factor field $\mathbb{Q}(\sqrt[3]{2})$ is not totally real: its two other embeddings are nonreal. A descent argument that requires total reality therefore cannot be applied to this factor. No field-descent theorem is used here.

The finite calculations are exact and reproducible. The complete mathematical argument is presented above, but neither independent human review nor formal proof-assistant verification has occurred. No priority claim relative to all existing literature is made. The accompanying independent-review report records the subsequent informal Codex AI-agent audit; no Lean verification was performed.

A A compact verification of the geometric argument

The following implication chain is useful for auditing the interface between the exact data and the unrestricted-width conclusion.

First, the arithmetic seed supplies seven rational three-dimensional subspaces U_i and an orthonormal basis $u_i \in U_i$. The form Q is positive semidefinite on each U_i , with kernel precisely $\mathbb{R}u_i$. The column rank-three certificates imply that every such line is irrational in the sense of Lemma 2.2.

Second, the triangle W supplies rational cones $G_i \subset U_i$ containing u_i with strictly positive coefficients. The 189 strict inequalities force positive bilinear pairings between nonzero vectors in different cones. Hence a zero of Q in their sum cannot use more than one group. This is the only step needed to pass from local semidefiniteness to the complete zero-set description.

Third, the common rational functional proves pointedness and the orthonormal basis proves full dimension. Exhaustive facet enumeration then identifies the sum cone *exactly* with $\{x : Rx \geq 0\}$. The inclusion of the basis gives $RO \geq 0$, while the equality of the two cone descriptions constrains *every* possible reduced factor column, not just the displayed seven-column factor.

Finally, rationality is preserved by the left inverse of the rational full-column-rank matrix R . The trace-zero identity turns the nonnegative quadratic values of every reduced factor column into individual zeros. The irrational-ray test then rules out every nonzero rational column, independently of how many columns there are.

None of these implications uses density of rational points to obtain an exact boundary factor, a conjectural rational exposing normal, or an assumption that rational cp-rank equals real cp-rank.

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